

## Repair Manuals

Timing chain: removal/installation

### Warnings and recommendations

Always use new gaskets

### Removal

Disconnect the battery earth cable

Place the bonnet in the service position

Remove the air inlet duct

Remove the air filter together with the mass airflow meter

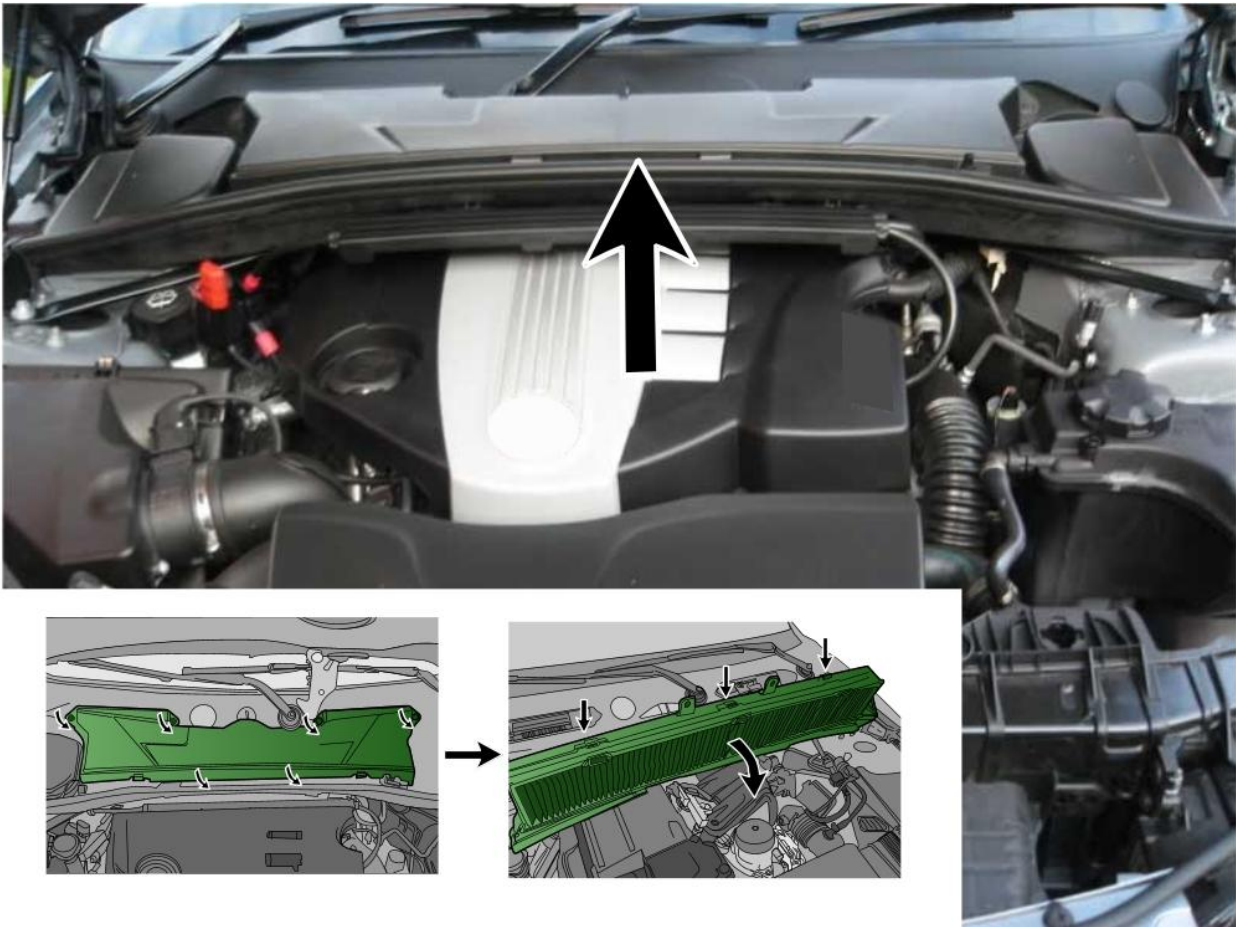
Remove the engine upper cover

Remove the ignition coils

Remove the spark plugs

Remove the dust and pollen filter

Dust and pollen filter location



Remove the engine side cover. On both sides.

Remove the dust and pollen filter housing

Disconnect the wiring loom from the valve cover

Disconnect the crankcase breather hose from the valve cover

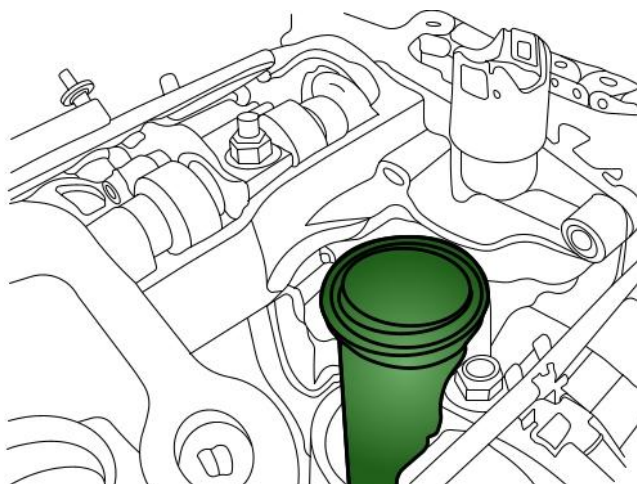
Loosen the bolts on the cylinder head cover. From outside to inside.

Remove the valve cover

Remove the spark plug housing from cylinders 1 and 4

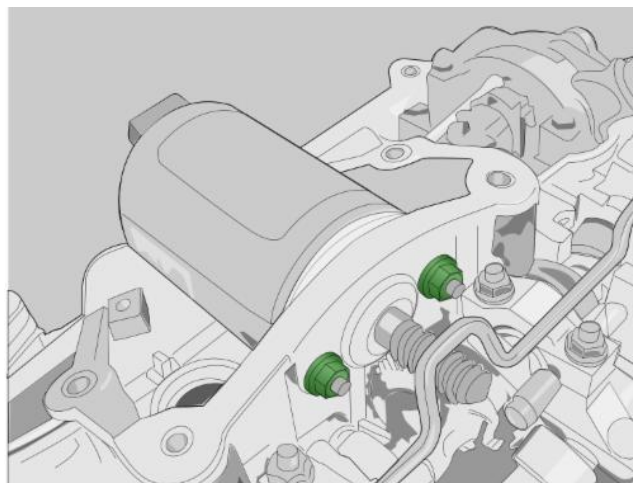
## Repair Manuals

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Renew the O-rings

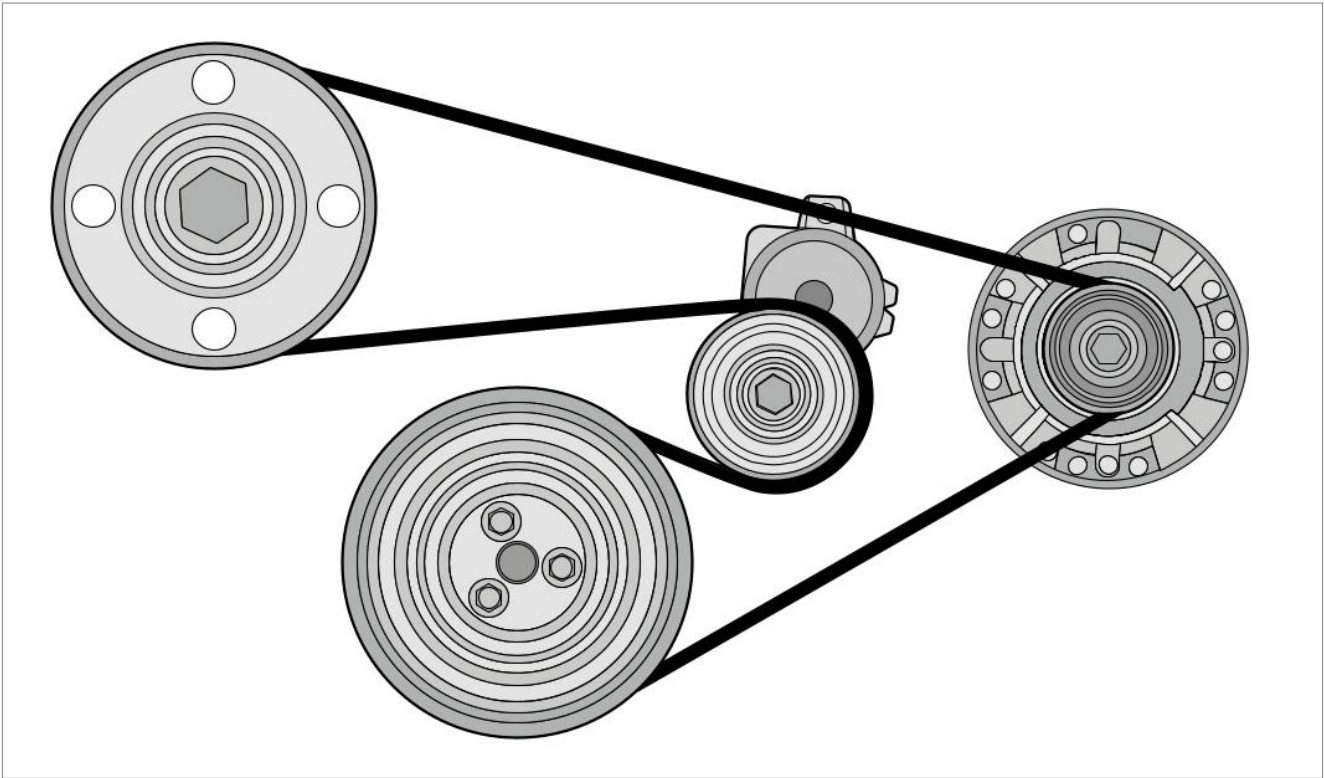
Remove the eccentric shaft actuator



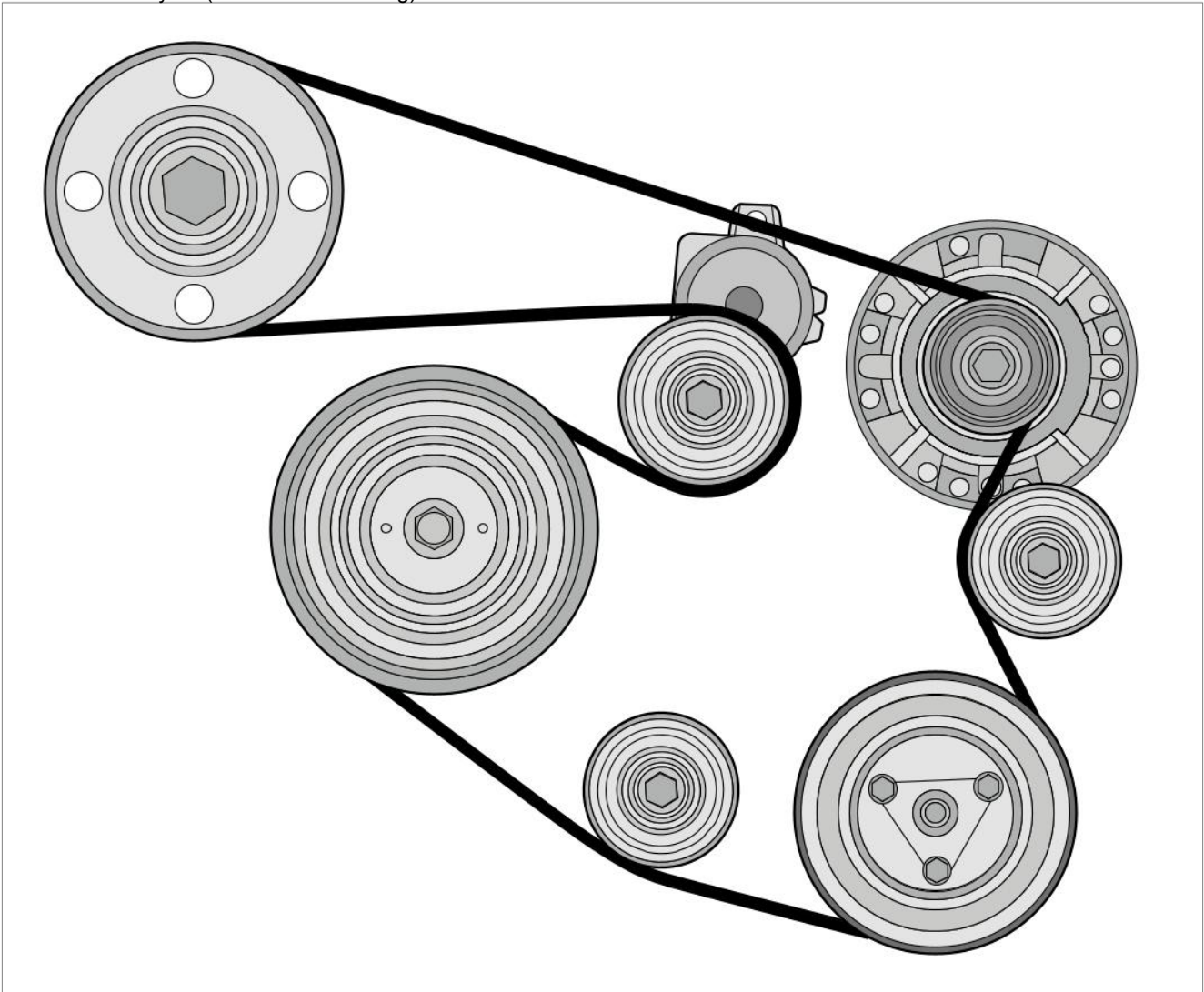
Remove the ancillary drive belt

Drive belt layout (Without air conditioning)

Repair Manuals



Drive belt layout (With air conditioning)

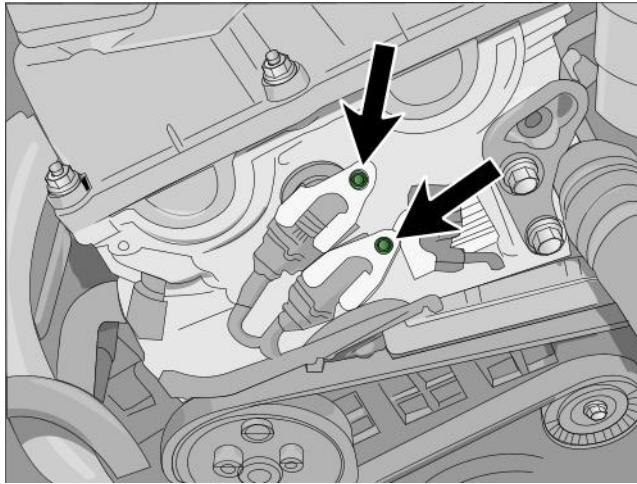


## Repair Manuals

Remove the crankshaft pulley

Disconnect the electrical connectors

Remove the valve timing control solenoid valves



Renew the O-rings

Remove the headlight support bracket

Support the engine with a jack stand (00 6 001, 00 6 031, 00 6 051, 00 6 052, 00 6 053)

Disconnect the engine supports, left and right

Raise the engine slightly ((10 mm))

Remove the engine lower covers

Disconnect the steering column from the steering box

Support the subframe (00 2 030, 31 5 250)

Remove the subframe bolts

Check the hoses

Lower the subframe ((< 10 cm))

Drain the engine oil

Disconnect the oil level sensor

Remove the oil sump

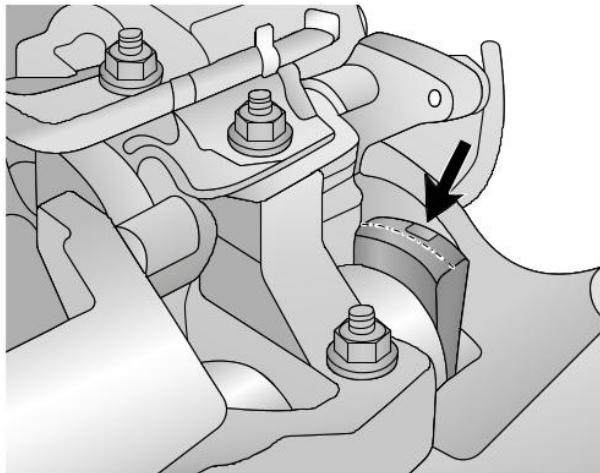
Refit the subframe

Remove the engine support tool (00 6 001, 00 6 031, 00 6 051, 00 6 052, 00 6 053)

Turn the crankshaft to TDC for cylinder 1

Check the position of the camshafts

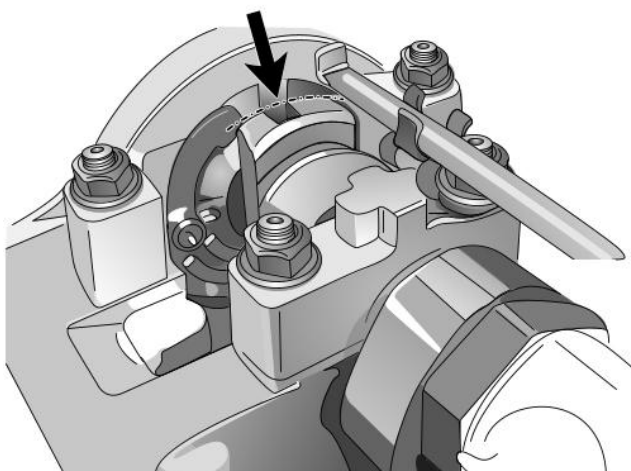
Inlet camshaft:



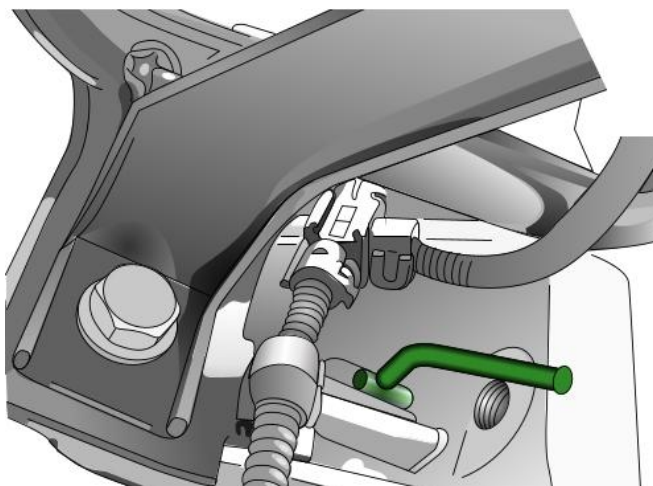
Exhaust camshaft:



## Repair Manuals

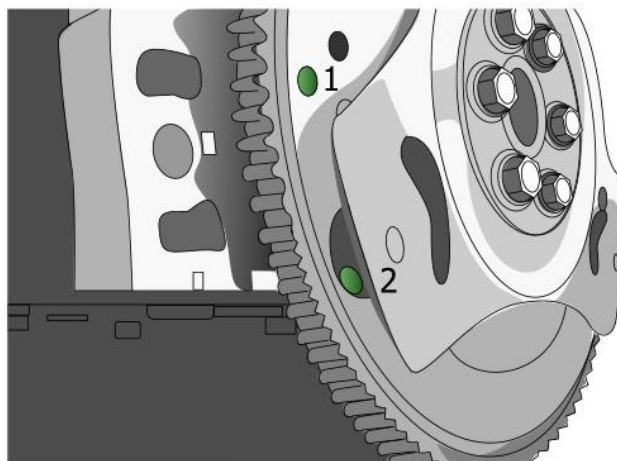


Lock the flywheel (11 5 120)



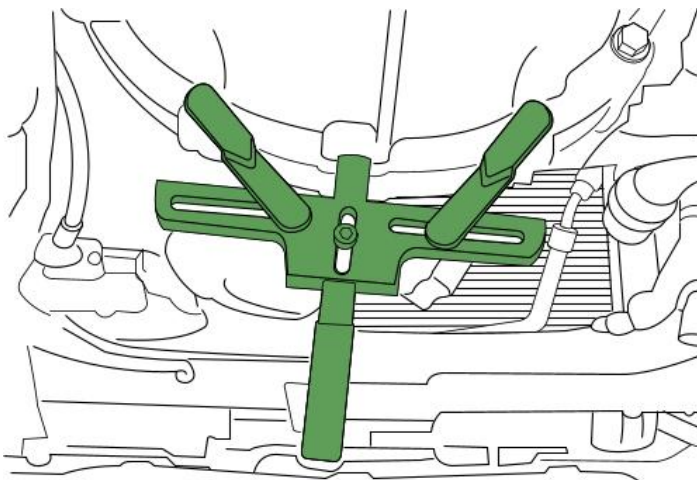
Vehicles with automatic transmission:

The special tool must be secured in bore No. 1



All vehicles:

Fit the crankshaft locking tool (11 8 180)

**Repair Manuals**

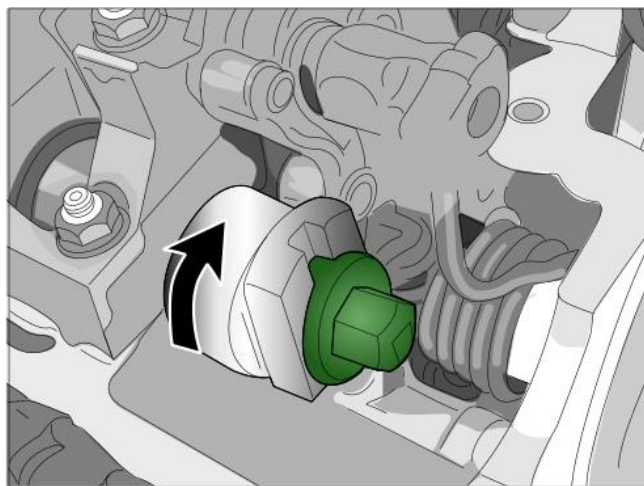
Loosen the crankshaft gearwheel centre bolt

Check the camshaft adjusters

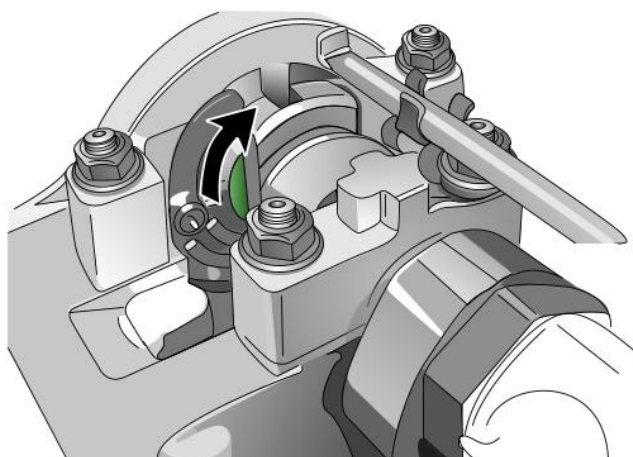
Turn the camshafts to check that the camshaft adjustment units can be locked

Use the special tool (11 9 270)

Inlet camshaft:



Exhaust camshaft:

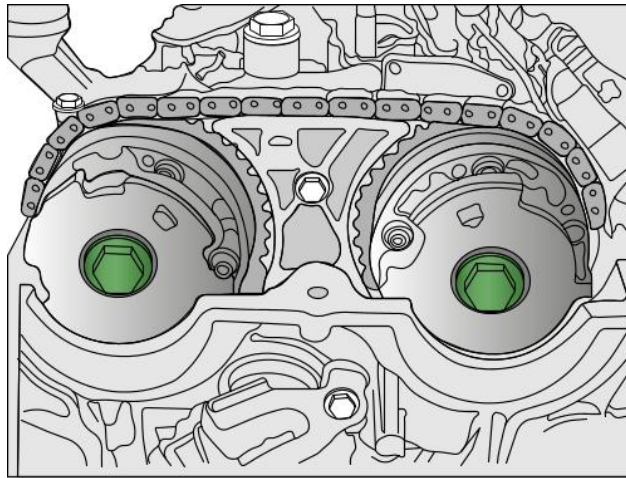


Action required if a fault is found:

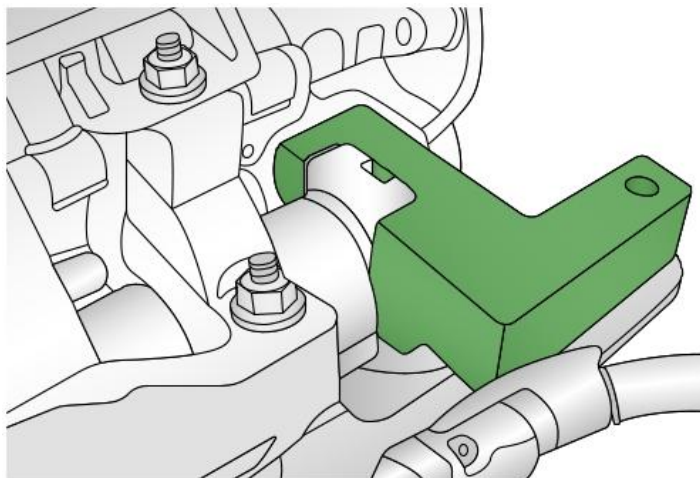
## Repair Manuals

Renew the camshaft gearwheels

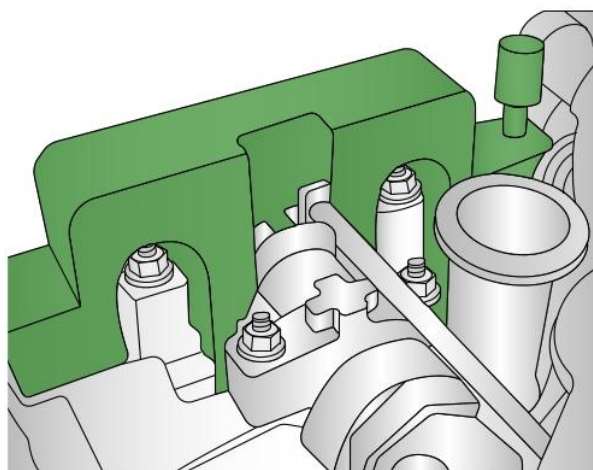
Loosen the camshaft gearwheel bolts



Fit the inlet camshaft locking tool (11 9 292)

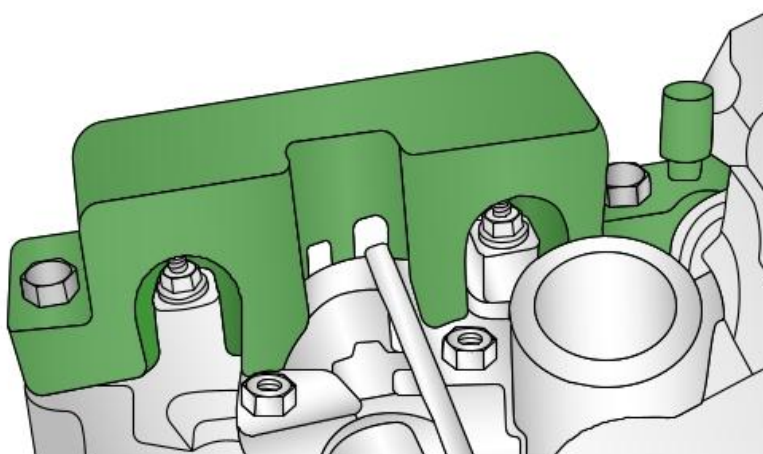


Fit the exhaust camshaft locking tool (11 9 291, 11 9 293)



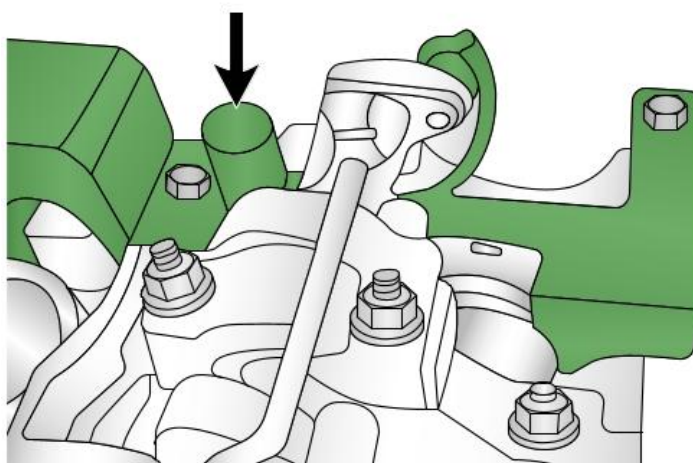
Secure the special tool with two bolts

## Repair Manuals

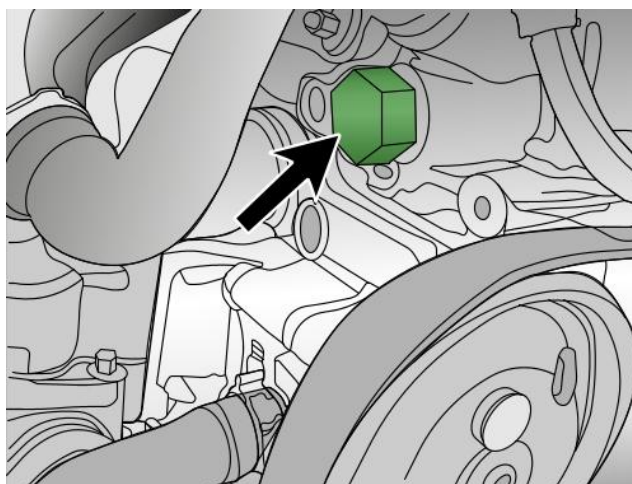


Tighten the special tool

Secure the special tool with a bolt (11 9 292)



Remove the chain tensioner

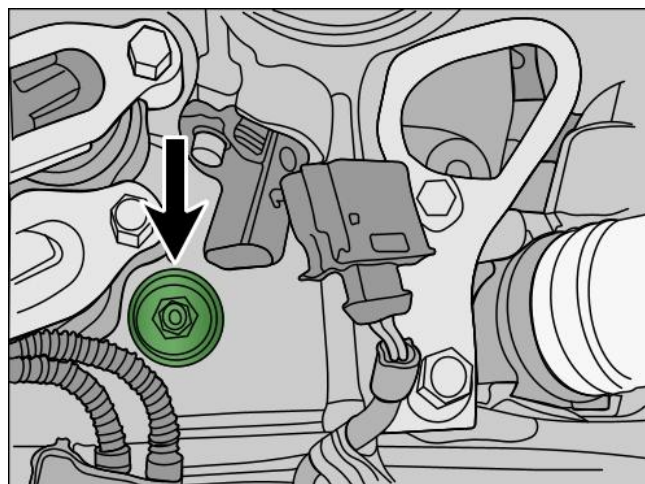
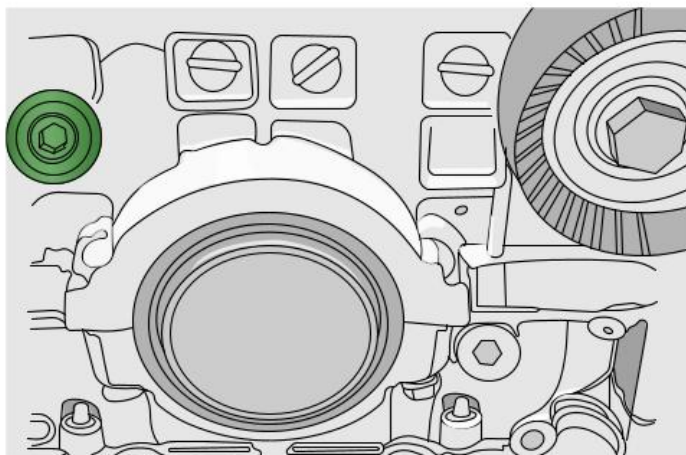


Remove the camshaft gearwheels

Remove the crankshaft hub

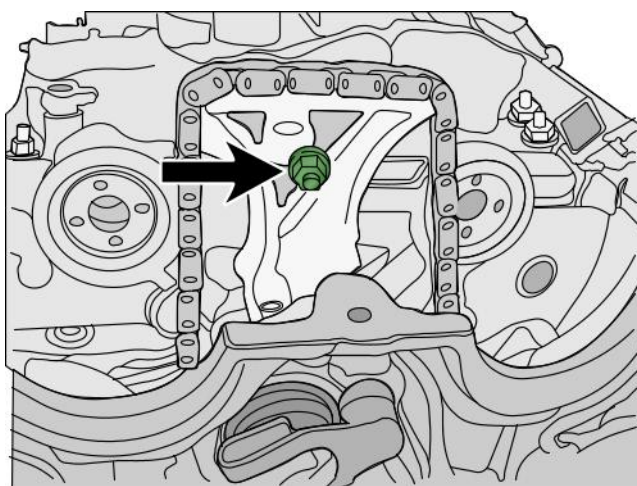
Remove the access plugs



**Repair Manuals**

Remove the bearing pins for the chain guide rail and the tensioner rail

Remove the bearing pin from the cylinder head



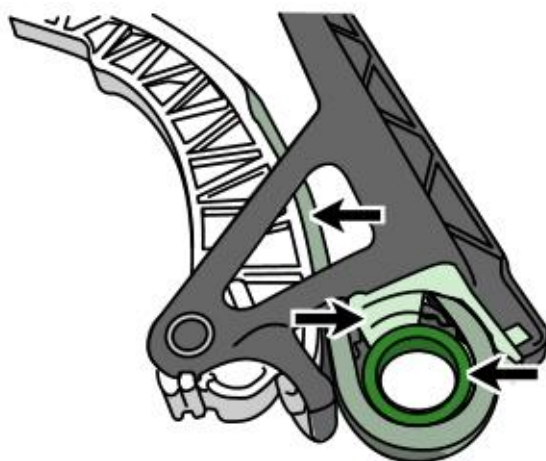
Remove the chain guide with the timing chain, guide rail and tensioner rail

**Installation**

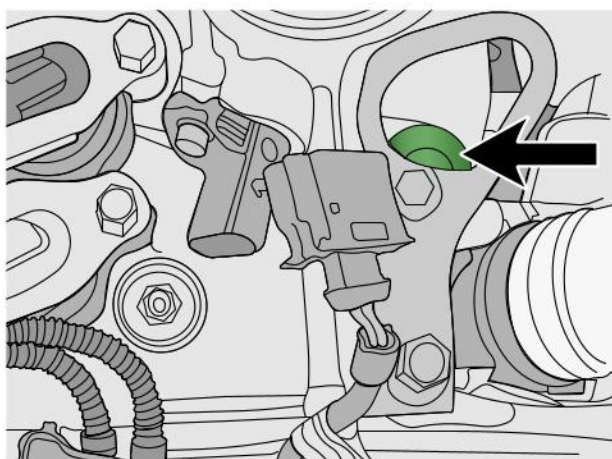
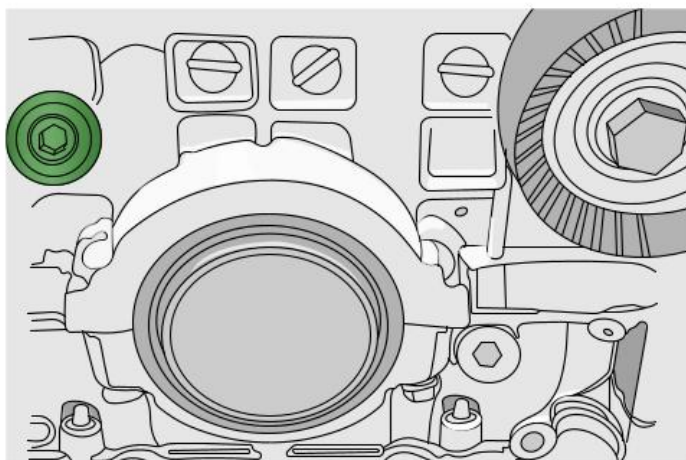
Refit the chain guide with the timing chain, guide rail and tensioner rail

Tension the chain

Note: Ensure correct fitting

**Repair Manuals**

Refit the bearing pins for the chain guide rail and the tensioner rail



Refit the access plugs

Refit the crankshaft hub

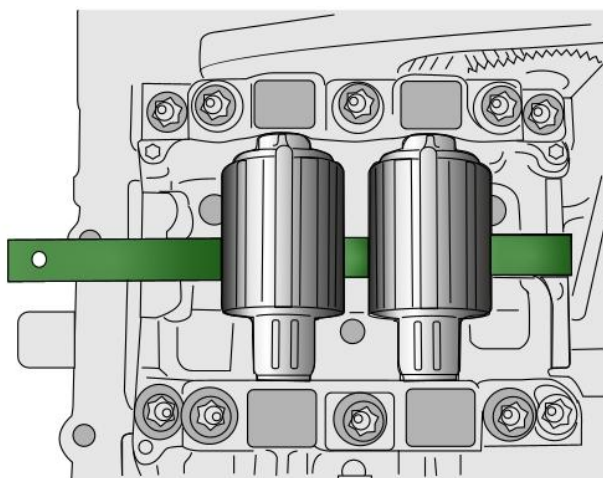
Check the condition and renew if necessary

Refit the crankshaft pulley centre bolt

Note: Finger-tight only

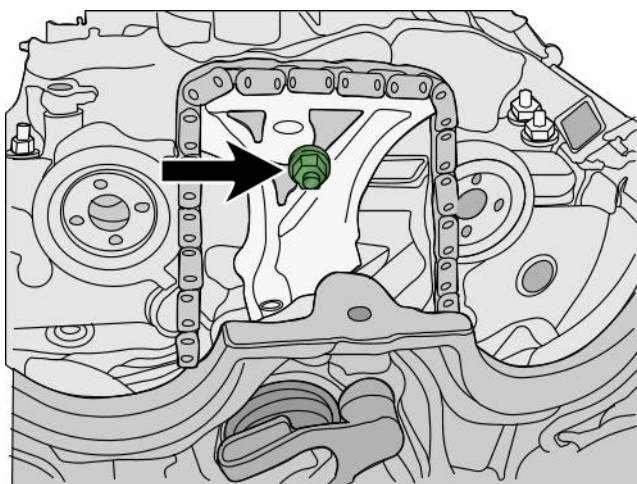
Align the balance shafts

Use the special tool (11 5 140)

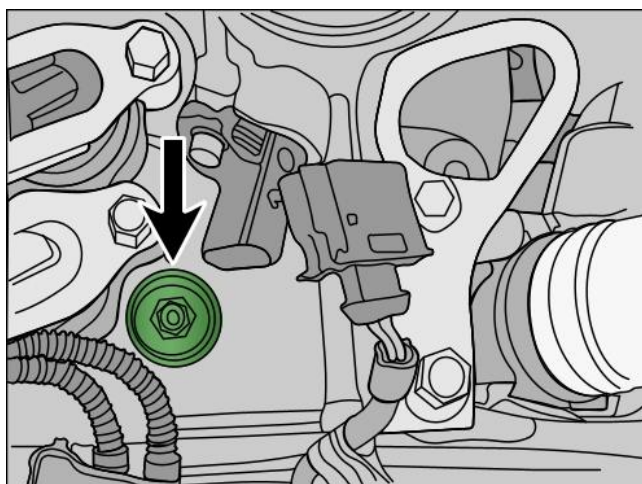
**Repair Manuals**

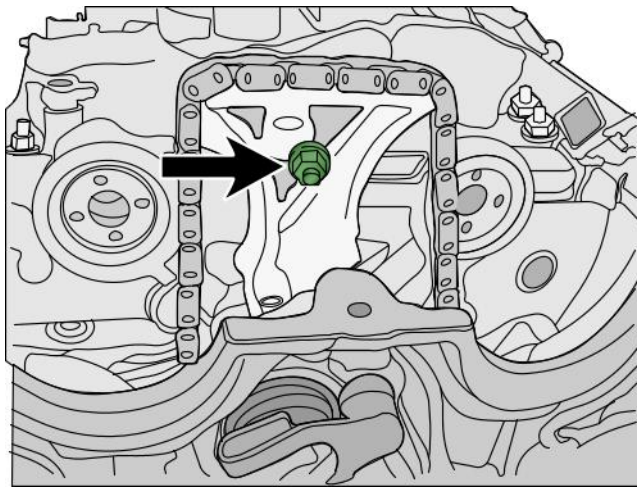
Refit the variable camshaft timing solenoids

Note: Fit the bolt finger-tight only



Tighten the bolts:



**Repair Manuals**

Check the camshaft gearwheel markings

Exhaust camshaft gearwheel: ((AUS/EX))

Inlet camshaft gearwheel: ((EIN/IN))

Pull the timing chain upwards

Tension the chain

Fit the timing chain onto the inlet camshaft gearwheel

Refit the camshaft sensor gear

Tighten the camshaft gearwheel bolt by hand

Note: Use a new bolt

Pull the timing chain upwards

Tension the chain

Fit the timing chain onto the exhaust camshaft gearwheel

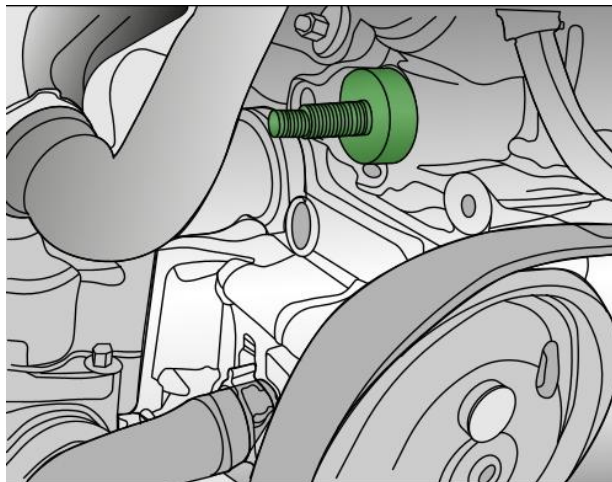
Refit the camshaft sensor gear

Tighten the camshaft gearwheel bolt by hand

Note: Use a new bolt

Press down on the tensioner rail

Fit the dummy chain tensioner (11 9 340)

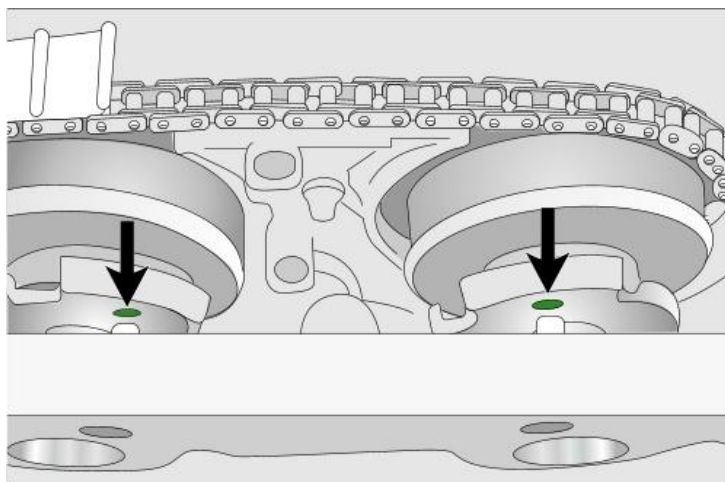


Fit the camshaft timing tool (11 9 350)

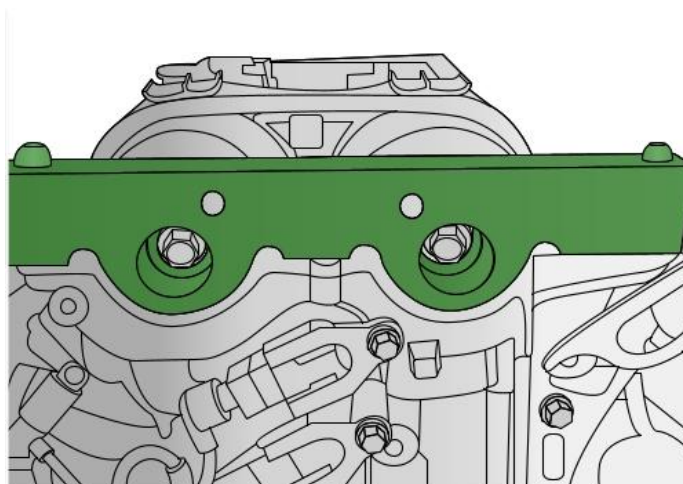
Align the camshaft gearwheel marks as shown



## Repair Manuals



Secure the special tool with two bolts (11 9 350)

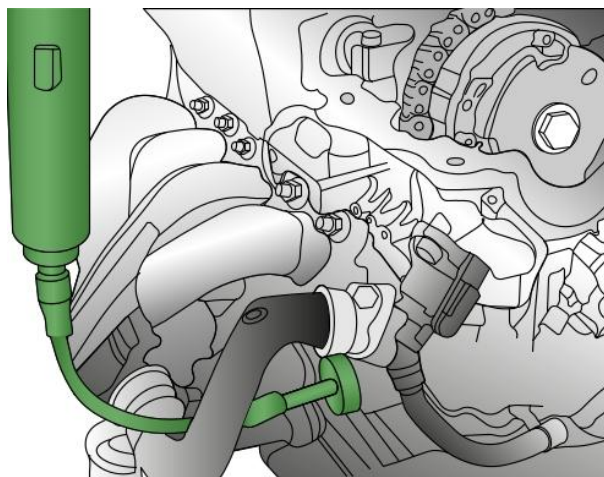


Loosen the camshaft gearwheel bolts ((180°))

Tighten the camshaft gearwheel bolts lightly until there is no play

Pre-tension the timing chain

Use the special tool ((0.6 Nm)) (00 9 250)

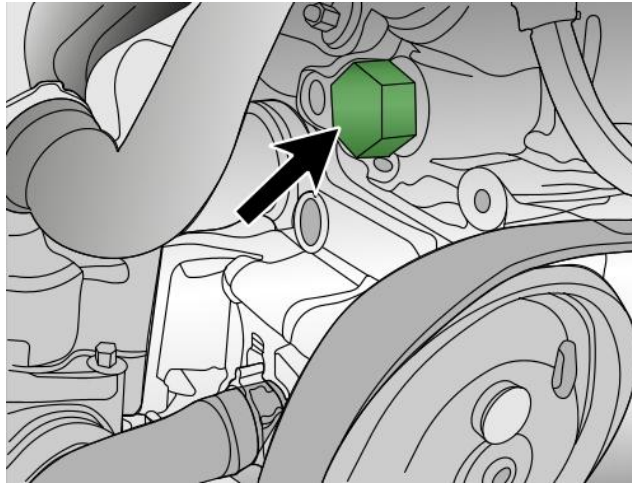


Tighten the camshaft gearwheel bolts (00 9 120)

Remove the special tools (11 9 340, 11 9 350)

Compress the chain tensioner piston twice to drain the oil

Refit the chain tensioner piston

**Repair Manuals**

Remove the camshaft locking tools (11 9 292, 11 9 291, 11 9 293)

Remove the flywheel locking tool (11 5 120)

Check that the crankshaft locking tool is in position (11 8 180)

Tighten the crankshaft pulley centre bolt

Remove the crankshaft locking tool (11 8 180)

Turn the crankshaft clockwise 2 rotations

Check the timing

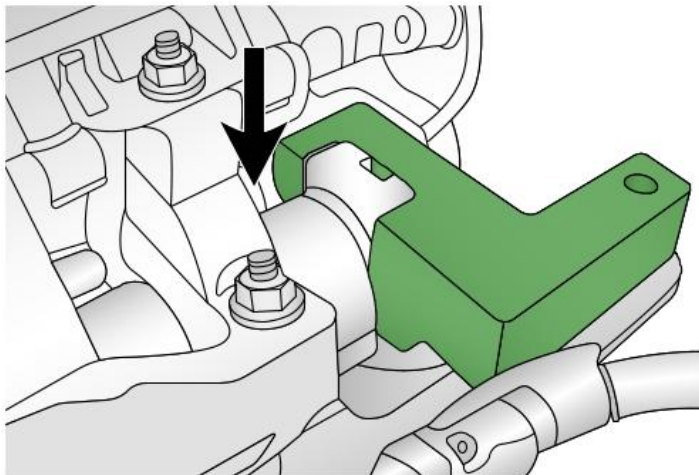
Turn the crankshaft to TDC for cylinder 1

Lock the flywheel (11 5 120)

Fit the inlet camshaft locking tool (11 9 292)

The special tool must be level with the cylinder head

Maximum protrusion ((0.5 mm))

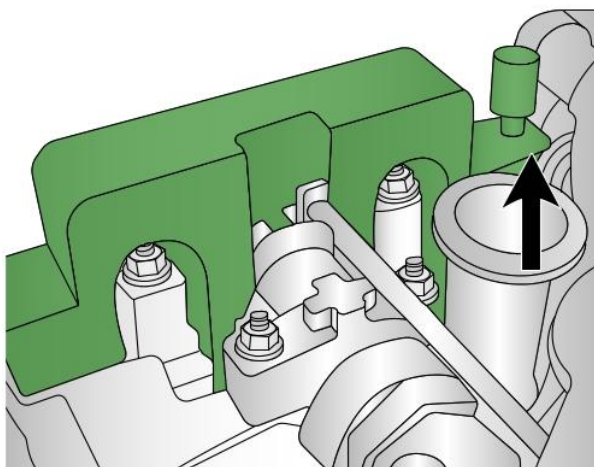


Fit the exhaust camshaft locking tool (11 9 291, 11 9 293)

The special tool must be level with the cylinder head

Maximum tolerance ((<1.0 mm))

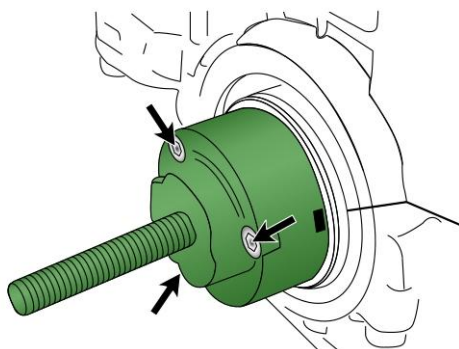
## Repair Manuals



The timing is set when the special tools fit correctly

Remove all the locking tools (11 5 120, 11 9 292, 11 9 291, 11 9 293)

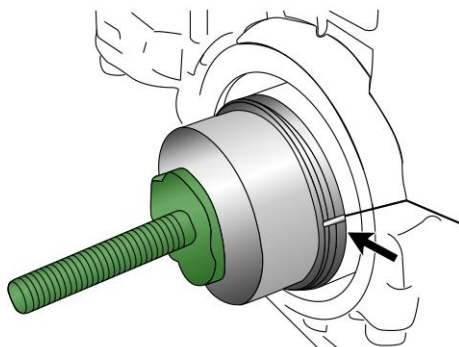
Fit the special tool (11 9 232)



Note: Clean off the sealing surfaces

Fit the crankshaft seal over the special tool

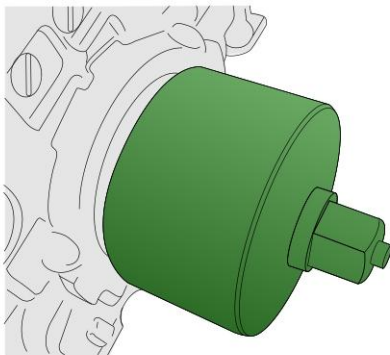
Note: The seal should be aligned with the timing cover



Refit the crankshaft front radial seal

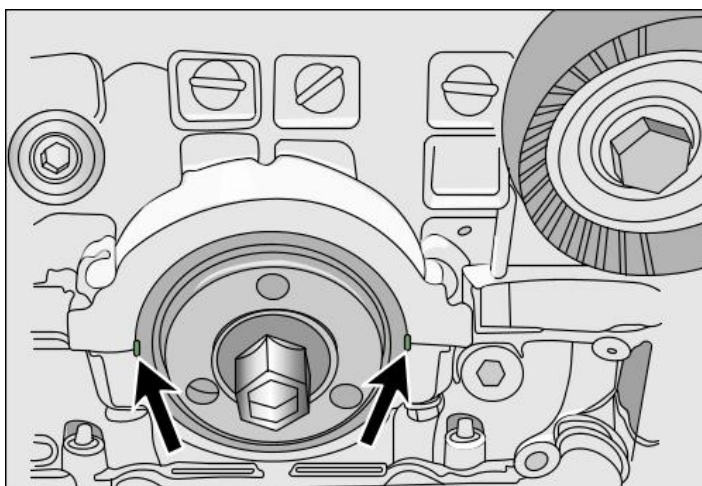
Use the special tool (11 9 231, 11 9 232, 11 9 233, 11 9 234)

## Repair Manuals



Remove the special tools (11 9 231, 11 9 232, 11 9 233, 11 9 234, 11 9 232)

Apply sealant as indicated. Loctite: ((171000)).



Further refitting is in the reverse order of removal

Refill the engine oil

Engine sump, including filter 4.25 (l)

Reconnect the battery earth cable

### Torque settings

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 180 (°)

Timing chain tensioner

Timing chain tensioner (Upper)

Renew the seal

Stage 1 65 (Nm)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt (Threads lightly oiled)

Stage 1 50 (Nm)

Stage 2 150 (°)

Oil sump



## Repair Manuals

Sump 30 (Nm)

Engine oil drain plug

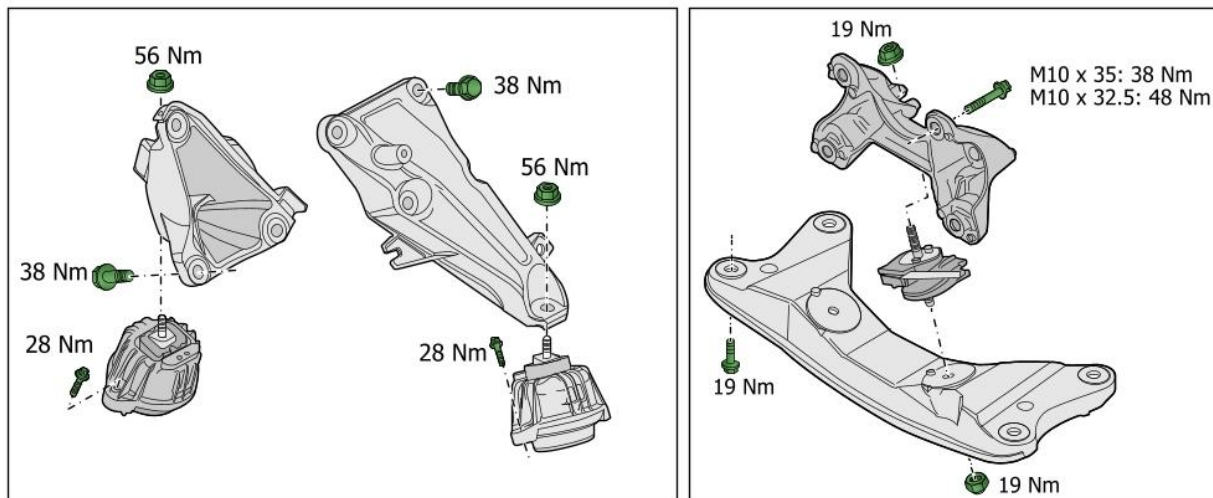
Engine oil drain plug 25 (Nm)

Suspension: Refer to 'Technical drawings'

Steering: Refer to 'Technical drawings'

Engine/gearbox mounts

Engine



Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Valve cover

Valve cover 10 (Nm)

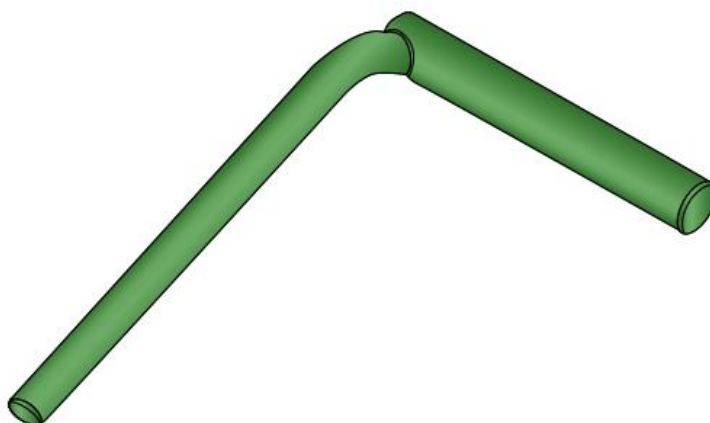
Spark plugs

Spark plugs (M12)  $23 \pm 3$  (Nm)

Spark plugs (M14)  $30 \pm 3$  (Nm)

### Special tools

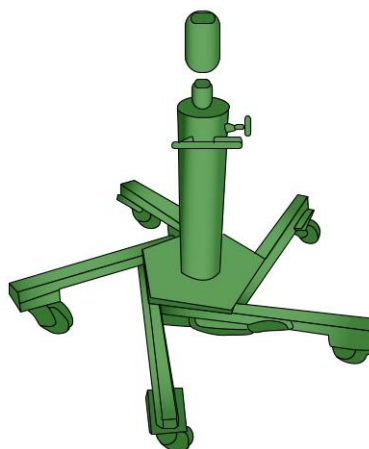
Flywheel locking tool: (11 5 120)



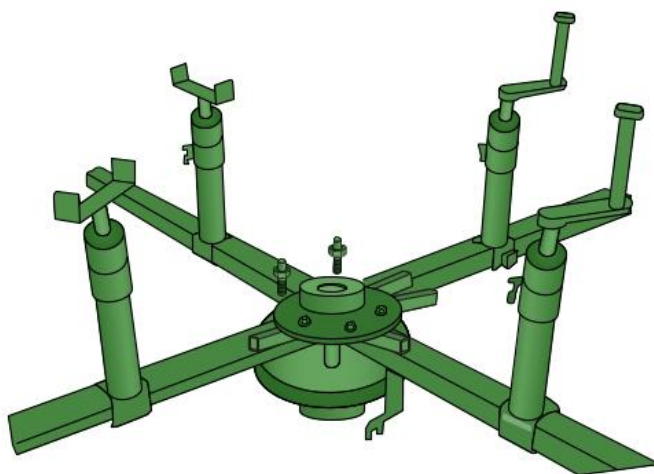
Transmission jack: (00 2 030)

## Repair Manuals

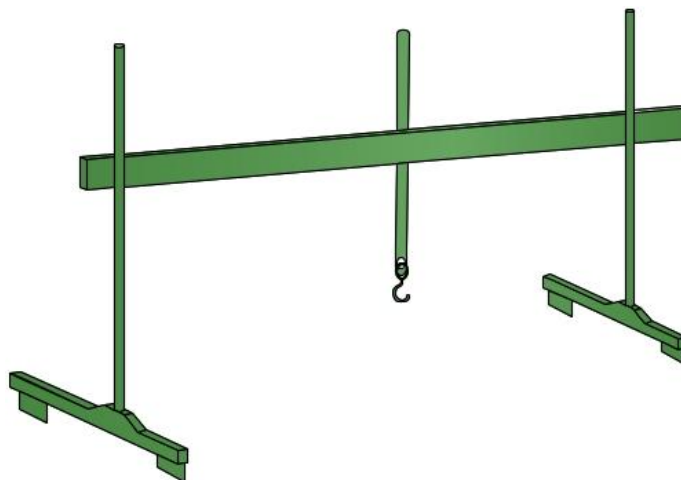
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Subframe support: (31 5 250)



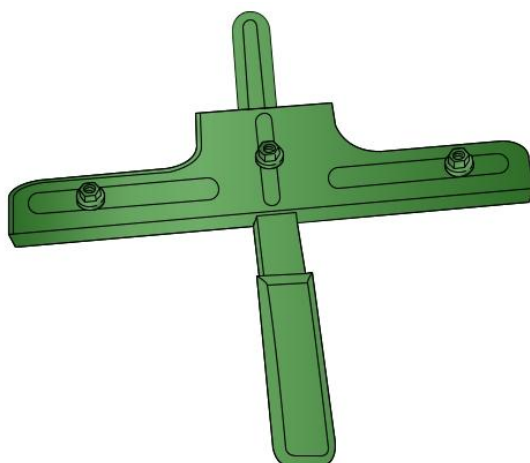
Engine support tool: (00 6 001, 00 6 031, 00 6 051, 00 6 052, 00 6 053)



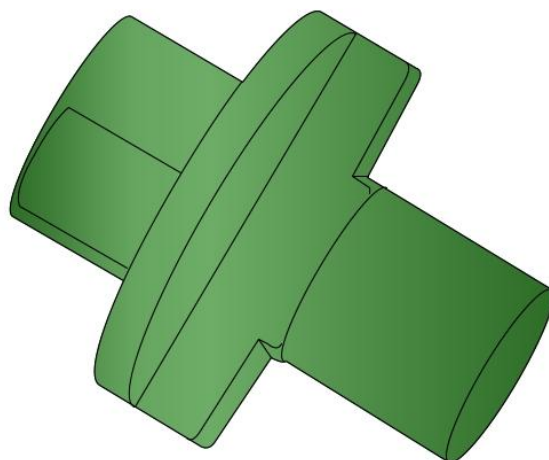
Crankshaft locking tool: (11 8 180)

## Repair Manuals

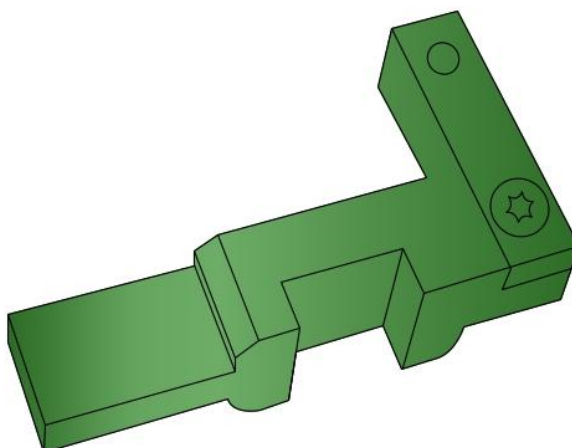
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Camshaft turning tool: (11 9 270)



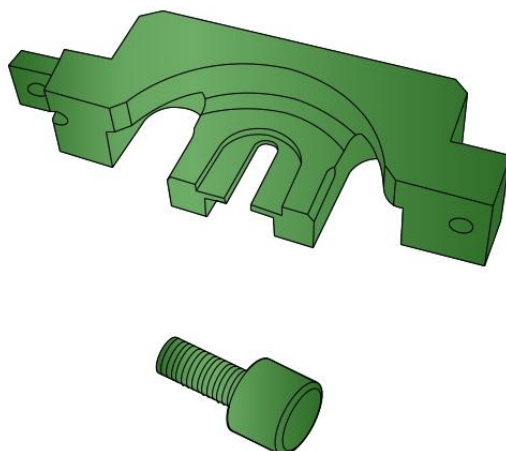
Camshaft locking tool: (11 9 292)



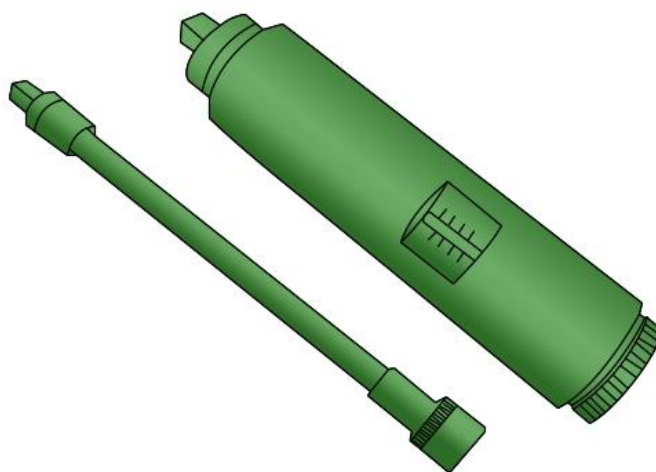
Camshaft locking tool: (11 9 291, 11 9 293)

## Repair Manuals

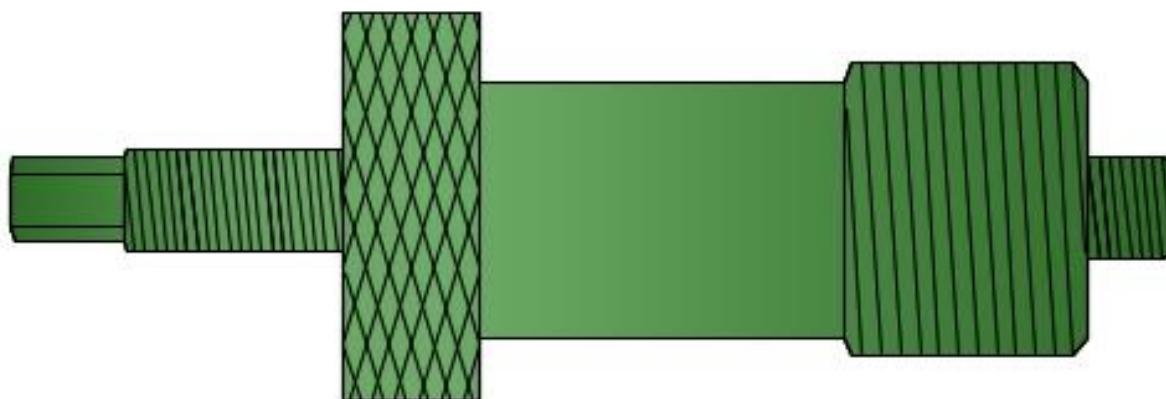
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Adjustment tool: (00 9 250)



Adaptor: (11 9 340)



Balance shaft tensioning tool: (11 5 140)

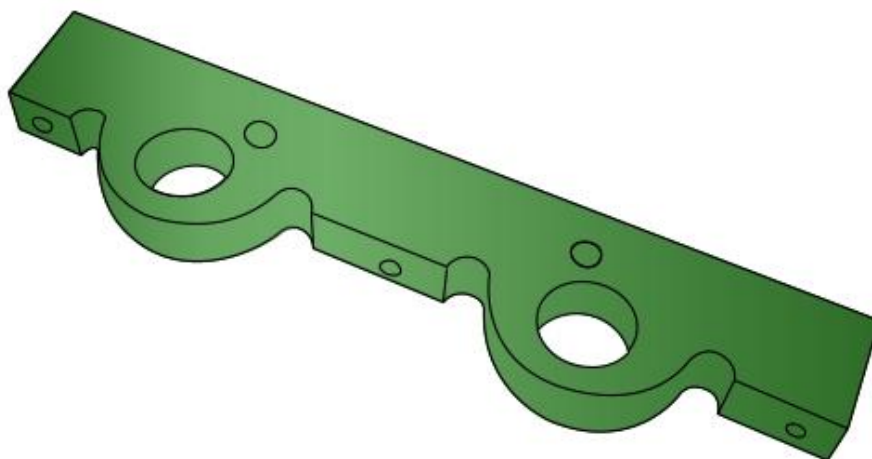


## Repair Manuals

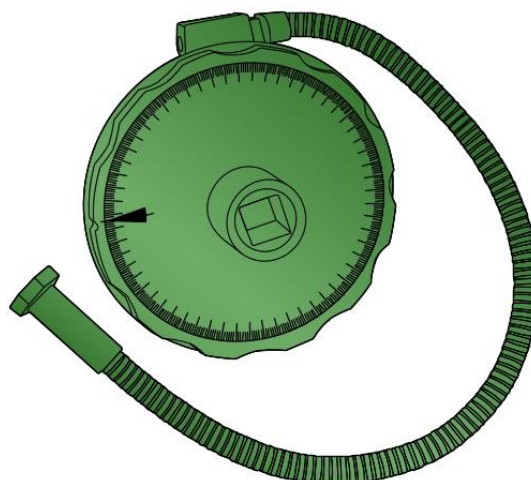
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Camshaft alignment tool: (11 9 350)



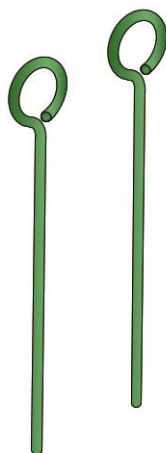
Torque wrench: (00 9 120)



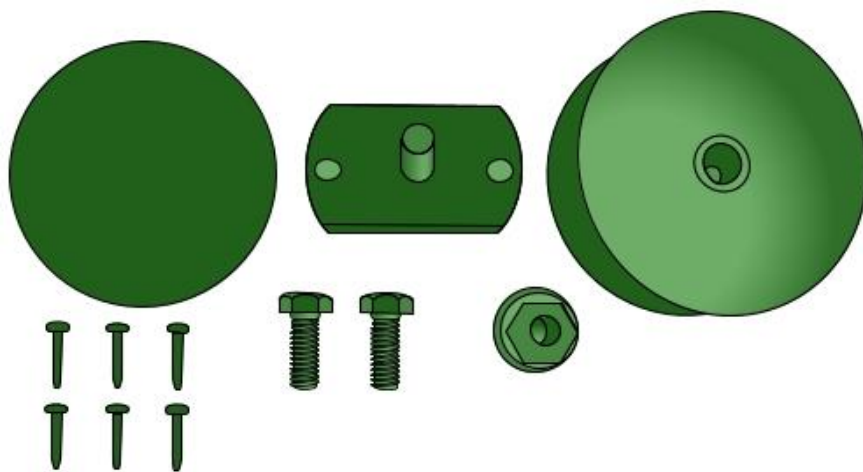
Chain tensioner locking pin: (11 3 340)

**Repair Manuals**

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Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)



Oil seal fitting tool: (11 9 232)

